

## Installation

Wireshark installation is very simple and straightforward.

**Windows:** Download the latest 32- or 64-bit version installable, which is compatible with your Windows version; double-click on it and follow the instructions.

During this installation, you will be asked to confirm installation of winpcap libraries. Here is an interesting point to remember for those paranoid about security: some of the websites, such as McAfee SiteAdvisor, rate winpcap libraries as security risk. This is primarily due to the ability of winpcap to capture network packets. So go ahead and install the libraries, which are a must to run Wireshark.

**Ubuntu:** Search for Wireshark under Ubuntu Software Centre, and click to install or use the command line:

```
sudo apt-get install wireshark
```

That's it. Your system is ready for the first capture. Welcome to the exciting world of network troubleshooting and protocol analysis.

## Capturing packets

**Windows:** Start Wireshark by clicking its icon.

**Linux:** Use the command `wireshark`. You may require administrative privileges. The recommended way to do so is to use `gksudo wireshark`.

Wireshark also has various command line options, which will be covered in a separate article. Optionally, you can capture packets using dumpcap (the packet capture engine), save them and use Wireshark for further analysis. This ensures minimum administrative privileges, limited only to the capture engine and not the entire Wireshark software.

Select the capture interface on the first screen. Go through Figure 1; did you notice something interesting under the Interface List? You can capture traffic on USB ports as well!

Click on the desired interface and Wireshark starts capturing and displaying packets, which are represented in three panes.

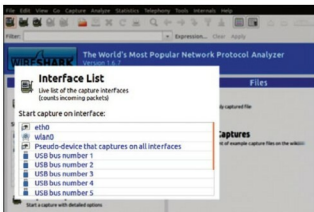


Figure 1: Wireshark interface selection

## The three Wireshark panes

The topmost pane is called the 'Packet List Pane' and displays the captured packet's number, its time stamps, source and destination addresses, protocol, length and other information about it.

The middle pane displays 'Packet details' of the current packet as viewed from various layers. The captured data could be viewed as *Frame – Physical, Ethernet – Data Link, IP – Network, UDP (or TCP) – Transport, and DNS (or any other application) – Application* layers. (For more details about these layers, there is ample material, including excellent videos, on the Web to get a better understanding.)

'Packet Bytes' are displayed in the bottom most pane.

If this sounds very simple, here is one interesting difficulty.

## Capturing packets in a switched environment

Capturing traffic towards and from Wireshark PC is simple enough; however, for network troubleshooting you require to capture various kinds of network traffic, such as what originates from any of the systems in the network, the entire network traffic from your network towards the Internet, etc.

Capturing traffic in a switched environment is not easy. And here's why. Let me start with the difference between an Ethernet hub and an Ethernet switch. The hub sends all packets received on a particular port to all the other ports. The switch forwards traffic only to the designated port. To achieve this, the switch maintains a table of MAC addresses (Layer 2 address corresponding to the port binding), and forwards the packet to the interface to which the device with the destination MAC address is connected. If the destination MAC is not known, it broadcasts to all ports (except the sender port), asking for the port that has the device with the destination MAC. Once a confirmatory reply is received, the actual packet is forwarded to the corresponding port only.

From a security perspective, an Ethernet switch is definitely better (than the hub). However, the feature ensuring security poses a challenge while capturing network traffic. In a switched network, the default capture will contain packets to and from the Wireshark machine and broadcast traffic only.

There are several ways to overcome this:

1. **Hubbing out:** Replace the switch with a hub and capture traffic by connecting the Wireshark system to the hub.
2. **Using Wiretap:** A 'tap' is used to capture communication between two end points. The primary concern while inserting Wiretap is that it should not disturb any traffic between two hosts. Wiretap achieves this by sitting between two end points (the switch and device) and providing two Ethernet outputs. These are connected to the Wireshark computer system with dual Ethernet. This system captures all traffic between two nodes without disturbing existing traffic. If this tap is inserted between the firewall and the switch, it will capture all Internet traffic. Typically, this box can be constructed by wiring four information outlets. See the wiring diagram for more details (Figure 2).